



Fig. 4: Geekbench score (Courtesy: primate labs.com/geekbench)

Go for an alternative architecture

We continue to see rapid growth of computational horsepower in mobile silicon enabling more and more advanced use cases.

"Smartphones and tablets are fast becoming our most frequently used computing devices for numerous activities (Web surfing, e-mail, social media, photography, casual gaming, etc), and we will see continued advancements in areas such as user-interface design and responsiveness, complex multitasking, realistic looking games, higher-quality screens and video processing, computational photography (such as HDR photos/videos using Tegra 4), improved context-sensitive speech processing, more intelligent location-aware computing, rich-media pattern recognition to quickly find/isolate video and audio information, and even complex transaction processing," adds Dhupar.

However, there seems to be an additional addition to silicon options. ARM-based SoCs are themselves modifiable and can be designed to a specific embedded application, and this is how firms like Samsung and Apple use their ARM licence to design and develop specific SoCs that have a little of the firm's own twists thrown in. Of course, this is not possible with Intel, which is where the trouble started for them. What if there were another technology that went one step further?

"Processor cores offered by ARM are not necessarily easy to modify themselves, even though the chip on

which they are used, i.e., system-on-chip, can be architected for a specific embedded application and the overall SoC better optimised than a standard chip from a company like Intel," adds another industry veteran.

He explains that some of the latest emerging technologies in processor and system design allow one or more processors and the entire SoC, or even an entire system of systems, to be optimised together for a specific application. Even more interesting is the fact that these technologies enable a whole new class of SoC architectures (hardware equivalent of software algorithms), called direct-connect architectures, to be automatically synthesised from high-level architectural models.

One such example can be seen in recent design platforms such as Sankhya Teraptor, which offers a platform for agile system development with models enabling Indian companies like Dreamchip Electronics to create SoCs optimised for specific applications such as tablets. These SoCs have already been announced, and use a proprietary multi-core 32-bit RISC and DSP processor architecture as well as a unique optimised direct-connect architecture.

So who wins?

Intel has already announced its x86-based Bay Trail atom processor, which is slated for launch in Q3 2013. Compared to Clover Trail, which runs on 32nm, Bay Trail is known to be built using 22nm lithography. So while Intel might not have the biggest market

share right now, its future depends on the performance of its next-generation chips.

ARM, on the other hand, currently holds a high ground, with most devices selling with ARM on the board.

"Intel offers processors, while ARM offers processor cores. That is the fundamental difference in their technologies that enables ARM to offer a product that can be integrated into a full SoC optimised for use in a specific application—a simple and important competitive advantage that Intel has tried but failed to overcome through expensive process-level improvements," explains a senior engineer from the field of chip design.

With even previously x86 focused firms like AMD planning to bring out ARM chips soon, ARM is surely on top right now.

There is also talk of high-end devices featuring x86, while the rest would rely on ARM. However, what we mean by high-end would decide finer details of choice.

"If we want high-end devices to perform telemetry and data transmission over the Internet for critical applications, x86 would still be equally qualified (or rather better). On the other hand, if high-end implies only video and audio capabilities, ARM is the straight choice. We have often talked about smartphones being used as mainstream computing devices in the future; if that starts happening in the near future, ARM could gain advantage over x86," adds Anand.

Or, we could see a new disruptive technology hitting the market if the underdog AMD finally starts producing both x86- and ARM-based processors, in addition to the cool GPUs that the ATI acquisition brought in. Or perhaps MIPS would edge in on the market since it also is an RISC-based architecture like ARM. Future prospects surely look interesting.

So with all this in mind, what chip will your next product be based on? ●

The author is a tech correspondent at EPY Bengaluru